

|                                                |                                             |
|------------------------------------------------|---------------------------------------------|
| Program : <b>Diploma in Textile Technology</b> |                                             |
| Course Code : <b>6068</b>                      | Course Title: <b>Textile Testing II Lab</b> |
| Semester : <b>6</b>                            | Credits: <b>1.5</b>                         |
| Course Category: <b>Program Core</b>           |                                             |
| Periods per week: <b>3 (L:0, T:0, P:3)</b>     | Periods per semester: <b>45</b>             |

### Course Objectives:

- To impart the practical knowledge of testing various textile materials such as yarn and fabric to determine its properties.

### Course Prerequisites:

| Topic                                                                       | Course code | Course name           | Semester |
|-----------------------------------------------------------------------------|-------------|-----------------------|----------|
| Testing of properties of textile materials such as fibre, yarn, and fabric. |             | Textile Testing       | 5        |
| Testing of properties of textile fibre                                      |             | Textile Testing I Lab | 5        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO n | Description                                                                            | Duration (Hours) | Cognitive level |
|------|----------------------------------------------------------------------------------------|------------------|-----------------|
| CO1  | Interpret the count, twist and strength of yarn.                                       | 11               | Applying        |
| CO2  | Identify the yarn quality parameters such as evenness, hairiness and classimat faults. | 11               | Applying        |
| CO3  | Execute different methods of measuring various fabric parameters.                      | 14               | Applying        |
| CO4  | Discuss the color matching of dyed fabric                                              | 6                | Understanding   |
|      | Lab Exam                                                                               | 3                |                 |

**CO-PO Mapping:**

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| <b>CO1</b>      | 3   |     |     | 3   |     |     |     |
| <b>CO2</b>      | 3   |     |     |     |     |     |     |
| <b>CO3</b>      | 3   |     |     | 3   |     |     |     |
| <b>CO4</b>      | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| Module Outcomes | Description                                                                                   | Duration (Hours) | Cognitive level |
|-----------------|-----------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Interpret the count, twist and strength of yarn.</b>                                       |                  |                 |
| M1.01           | Determine yarn count by digital balance                                                       | 1                | Applying        |
| M1.02           | Determine the TPI of yarn by tension type twist tester                                        | 2                | Applying        |
| M1.03           | Determine the TPI of yarn by take - up twist tester                                           | 2                | Applying        |
| M1.04           | Study of Single yarn strength testing by Uster Tensorapid.                                    | 3                | Applying        |
| M1.05           | Determine of Yarn CSP by lea strength tester                                                  | 3                | Applying        |
| <b>CO2</b>      | <b>Identify the yarn quality parameters such as evenness, hairiness and classimat faults.</b> |                  |                 |
| M2.01           | Study of yarn evenness by ASTM black board appearance method.                                 | 2                | Understanding   |
| M2.02           | Study of Yarn evenness testing by Uster evenness tester                                       | 3                | Understanding   |
| M2.03           | Study Yarn hairiness testing by Zweigle's hairiness tester                                    | 3                | Understanding   |
| M2.04           | Study of Classimat faults.                                                                    | 3                | Understanding   |
|                 | Lab Exam - I                                                                                  | 1.5              |                 |
| <b>CO3</b>      | <b>Execute different methods of measuring various fabric parameters.</b>                      |                  |                 |
| M3.01           | Determine the count of yarn in the fabric by Beesley's balance.                               | 2                | Applying        |
| M3.02           | Measure the thickness of fabric by Fabric thickness tester                                    | 1                | Applying        |

|            |                                                                                                 |     |               |
|------------|-------------------------------------------------------------------------------------------------|-----|---------------|
| M3.03      | Determine the crimp % in yarn by Shirley yarn crimp tester.                                     | 1   | Applying      |
| M3.04      | Determination of pilling in fabric by Martindale abrasion tester.                               | 1   | Applying      |
| M3.05      | Study the method of measuring Fabric tensile strength by CRT principle.                         | 1   | Applying      |
| M3.06      | Determine the tearing strength of given fabric sample by Elmendorf Tearing strength tester      | 1   | Applying      |
| M3.07      | Determine the bursting strength of given fabric sample by fabric bursting strength tester       | 2   | Applying      |
| M3.08      | Determine the Fabric stiffness by Shirley stiffness tester                                      | 1   | Applying      |
| M3.09      | Measure the crease recovery angle of the given fabric sample by Shirley crease recovery tester. | 1   | Applying      |
| M3.10      | Study of determining abrasion resistance by Martindale abrasion tester.                         | 1   | Understanding |
| M3.11      | Determine the drape coefficient of the given fabric sample using drape meter.                   | 1   | Applying      |
| M3.12      | Study of Fabric air permeability tester                                                         | 1   | Understanding |
| <b>CO4</b> | <b>Discuss the color matching of dyed fabric</b>                                                |     |               |
| M4.01      | Perform the Colour matching of dyed fabric                                                      | 2   | Applying      |
| M4.02      | Study the method of measuring Light fastness of dyed materials.                                 | 1   | Understanding |
| M4.03      | Study the method of measuring Washing fastness of dyed materials.                               | 1   | Understanding |
| M4.04      | Study the method of measuring Perspiration fastness of dyed materials.                          | 2   | Understanding |
|            | Lab Exam - II                                                                                   | 1.5 |               |

The students should keep record for wet processing with samples and submit certified & bonafide record at the time of practical examination.

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                 |
|------------|--------------------------------------------------------------------------|
| T1         | J.E. BOOTH - Principles of Textile Testing (Butterworth third edition)   |
| R2         | Grover - Hand book of Quality control and Textile testing Butterworth    |
| R3         | Arindam Basu- Textile Testing Fibre,yarn, fabric SITRA publication       |
| R4         | Alagappan & Gopalakrishnan - Textile Testing SSM institute of technology |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.astm.org/Standards/textile-standards.html">https://www.astm.org/Standards/textile-standards.html</a>                  |
| 2            | <a href="https://www.testertextile.com">https://www.testertextile.com</a>                                                                  |
| 3            | <a href="http://www.uster.com/en/instruments/fiber-testing/uster-afis-pro/">www.uster.com/en/instruments/fiber-testing/uster-afis-pro/</a> |
| 4            | <a href="http://www.youtube.com">www.youtube.com</a>                                                                                       |